

Introduction to Hacking

What Is Hacking? - Hacking is the process of identifying and exploiting weakness in a system or a network to gain unauthorized access to data and system resources by compromising the security.

Example of Hacking: Exploiting the weakness of default password to gain access to the data stored inside the system.

What is Ethical Hacking? -

Ethical Hacking sometimes called as Penetration Testing is an act of intruding/penetrating into system or networks to find out threats, vulnerabilities in those systems which a malicious attacker may find and exploit causing loss of data, financial loss or other major damages.

What is the Purpose of Ethical Hacking? - The purpose of ethical hacking is to improve the security of the network or systems by fixing the vulnerabilities found during testing.

Who is an Ethical Hacker - An Ethical Hacker is a skilled professional who has excellent technical knowledge and skills and knows how to identify and exploit vulnerabilities in target systems. He works with the permission of the owners of systems.

Types of Hackers -

1. Black Hat Hackers - Black-hat Hackers are also known as an Unethical Hacker or a Cracker. These people hack the system illegally to steal



money or to achieve their own illegal goals. They find banks or other companies with weak security and steal money or credit card information. They can also modify or destroy the data as well. Black hat hacking is illegal.

2. White Hat Hackers - White Hat hackers are also known as Ethical Hackers. They never intent to harm a system, rather they try to find out weaknesses in a computer or a network system as a part of penetration testing and vulnerability assessments. Companies hire them to test their system.

3. Grey Hat Hackers - Grey hat hackers are a blend of both black hat and white hat hackers. They act without malicious intent but for their fun, they exploit a security weakness in a computer system or network without the owner's permission or knowledge. Their intent is to bring the weakness to the attention of the owners and getting appreciation or a little bounty from the owners.

4. Suicide Hackers - Suicide hackers are those who aim to bring down critical infrastructure for a "cause" and are not worried about facing jail terms or any other kind of punishment. Suicide hackers are similiar to suicide bombers who sacrifice their life for an attack and are thus not concerned with the consequences of their actions.

5. Script Kiddies - Script Kiddies are amateurs in the field of hacking. They try to hack the system with scripts from other fellow hackers. They try to hack the systems, networks, or websites. The intention behind the hacking is just to get attention from their peers. Script Kiddies do not have complete knowledge of the hacking process.

6. State-Sponsored/Country-Sponsored Hackers - State-sponsored hackers are those individuals employed by the government to penetrate, gain top-secret information

and damage the information systems of other governments.

7. Hactivist - Hactivist breaks into government or corporate computer systems as an act of protest.

In general, most hacktivism involves website defacement or denial of-service attacks.

Rules for Ethical Hacker :

- 1. Get a written permission from the owner/organization before performing the hacking.**
- 2. Maintain the privacy of that organization.**
- 3. Transparently report to the owner/organization about the weakness.**
- 4. Inform the related issue to the software and hardware vendor.**

Type of Cyber Crime :

- 1. Spam - Fake Mailing System etc.**
- 2. Privacy Violation - Exposing the data over the internet.**
- 3. Identity Theft - Stealing information from the system/network**
- 4. Sharing Copyrighted files/informtion - leaking the confidential files**
- 5. Eletronic Money Transfer - Net Banking - Transaction Procedure**
- 6. Electronic Money Laundering - Convert Black money to white**
- 7. ATM Fraud - Cloning of atm card**
- 8. DOS Attack - Denial of Service Attack (Router/ Server)**
- 9. MITM Attack - Man in the Middle Attack (Capturing packets)**

TYPES OF INFORMATION :

- 1. Confidentials Information - Aadhar Cards, Votar card, Birth Certificates, PAN Cards etc.**
- 2. Financial Information - Financial Statements, Bank Details, Login Credentials for banking poractices etc.**
- 3. Health Information - Policies, Diseases etc.**
- 4. Personal Information - Address, Phone Numbers, DOBs etc.**

Motive for hacking :

- 1.Hobby**
- 2.Defame/Revenge**
- 3.Stealing information**
- 4.Leaking information**
- 5.Security Reasons(Penetration Testing)**
- 6.Just For Fun**

Essential Terminology :

Hack value

It is a notion among hackers that something is worth doing or is interesting.

Vulnerability

Existence of a weakness, design, or Implementation error that can lead to an unexpected event compromising.

Payload

Payload is the part of an exploit code that performs the intended malicious action, such as destroying creating backdoors, and hijacking computer.

Exploit

A Breach of IT system through vulnerabilities.

Zero-Day Attack

An attack that exploits computer application vulnerabilities before the software developer releases a patch for the vulnerability. Basically they don't have patch of this vulnerability, hackers can exploit this vulnerability regularly.



Some Important Characteristics Of Information Security -

CIA Triad = Confidentiality - Integrity - Availability - A concept that focuses on the balance between the confidentiality, integrity and availability of data under the protection of your in

information security program.

Confidentiality - Confidentiality ensures that an Information is accessible to only an authorized user. The main purpose of confidentiality is to protect the sensitive information from reaching the wrong hands. It is used to maintain the privacy of the people. Encryption is a good example of confidentiality.

Integrity - Integrity maintains the correctness or accuracy of the information while the data is in transit, storage or processing. It is the guarantee that information is trust worthy and not tampered. This attribute ensures that an unauthorised person will not be able to modify the data.

RSA digital signature, SHA1 hash codes are good examples.

Availability - Information should be available to an authorised person when it is requested for. It is the guarantee of access to the authorised individual to information. Keeping all the hardware and software up to date and keeping back up, taking proper recovery measures will ensure availability of data.

